

Personal Learning Journey

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Data Understanding Preprocessing and Quality Assurance

Project context. Our group analysed the UCI Online Retail dataset to identify customer segments and segment-specific product associations for retention, reactivation and cross-selling decisions. This reflection explains my individual contribution, learning and development during the project.

My contribution

My main responsibility was to understand the UCI Online Retail dataset and prepare reliable inputs for the other members. I verified the raw file, documented its eight variables, explained that each row represented an invoice line, and investigated missing values, duplicate rows, cancellations, returns, invalid prices, administrative StockCodes and geographic imbalance. I then created task-specific processed datasets instead of forcing one cleaned dataset to serve every analysis. This work produced 391,150 customer-linked merchandise rows for segmentation and a broader 522,568-row merchandise basket dataset for association mining. I also recorded preprocessing decisions, row-count audits, data fingerprints and handover checks.

What I learned

The most important lesson was that data cleaning is not simply deleting unusual records. Repeated InvoiceNo values were valid because one invoice contains many products, while exact duplicate rows could inflate revenue and frequency. Missing CustomerID values could not be safely imputed for customer segmentation, but some anonymous transactions were still useful for overall basket analysis. I learned to connect every cleaning rule to the unit of analysis and the business question. I also improved my understanding of data leakage, reproducibility, identifiers, positive-purchase definitions and the difference between raw evidence and analytical datasets.

Challenge and judgement

My main challenge was deciding how to treat extreme values and administrative transactions. Large purchases could be genuine wholesale behaviour, so automatically deleting IQR outliers would have removed valuable customers. I retained them and recommended log transformation, scaling and stability testing downstream. Administrative charges were excluded from merchandise behaviour but preserved for sensitivity analysis. This taught me to avoid convenient rules that create unjustified information loss. I became more confident in defending a decision with counts, business meaning and downstream impact rather than saying that a method is standard practice.

Teamwork and future development

My outputs became the foundation for feature engineering, clustering and association mining, so clear handovers were essential. I used assertions, reload validation and SHA-256 fingerprints to help the team reproduce identical inputs. Reviewing later notebooks showed me how an early data-quality decision changes cluster profiles and rule support. Generative AI helped me troubleshoot code and improve explanations, but I verified suggestions using notebook results and the decision log. In future work, I would add automated schema tests, a UK-only sensitivity analysis and clearer pipeline functions. Overall, I learned that dependable machine learning begins with transparent data reasoning, not with the final algorithm.